Week 6 Notes

C-GAN

A standard GAN is like a talented artist who paints completely at random; you hit "run" and get a surprise mixture of shoes, shirts, or dresses with zero control over the outcome. A Conditional GAN (C-GAN) fixes this by adding a label vector directly to the input, handing the network a specific order like "make a coat." This changes the entire pipeline: the generator mixes random noise with the class tag to follow a concrete blueprint, while the discriminator gets a double checkpoint, checking not just if the image looks real, but if it actually matches the requested tag. By tying the mathematical framework to these explicit categories, you turn a chaotic guessing game into a predictable tool that generates exactly what you ask for on command.

Adversarial Auto-Encoders

An Adversarial Autoencoder (AAE) combines a standard autoencoder with a GAN to map data into a well-behaved latent space without the rigid mathematical constraints of a VAE. Instead of relying on volatile Kullback-Leibler divergence calculations that force the use of a Gaussian distribution, an AAE uses an adversarial game to minimize the highly stable Jensen-Shannon Divergence. This flexibility allows you to shape your latent space using absolutely any target prior distribution you want, including uniform or discrete layouts.

The architecture trains in two distinct steps. First, a reconstruction phase forces the encoder and decoder to work together to compress and accurately recreate input images. Second, a regularization phase introduces a discriminator to look at latent vectors. It tries to distinguish between true samples drawn directly from the chosen prior distribution and fake vectors produced by the encoder. Meanwhile, the encoder acts as a generator, constantly updating its weights to fool the discriminator. Once this balance is reached, the encoder can be removed completely; you simply sample raw noise from the target prior and feed it straight into the decoder to synthesize entirely new images.